Day 6 - APTITUDE

Block - 1
(Permutation and Combination.

Question - 1
In how many ways can 3 students be selected from 7 students?

Sol :-

$$^7C_3 = \frac{7!}{3! \times 4!}$$

$$= \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1)(4 \times 3 \times 2 \times 1)}$$

$$\Rightarrow \frac{7 \times 6 \times 5}{3 \times 2 \times 1} \quad \boxed{\Rightarrow 35} \Rightarrow \text{Answer.}$$

2) How many 2-digit numbers can be formed using digits 1, 2, 3, 4 without repetition? Digits = {1, 2, 3, 4}

Sol :- To find? How many ways to form 2-digits numbers

First digit ⇒ 4 choices
Second digit ⇒ 3 choices
⇒ 4 × 3 ⇒ 12

$$\Rightarrow {}^4P_2 = \frac{4!}{2!} \quad \Rightarrow \frac{4 \times 3 \times 2 \times 1}{2 \times 1} \quad \boxed{\Rightarrow 12}$$

$\boxed{12} \Rightarrow$ Answer

3) In How many ways can a captain and Vice-captain be choosen from 6 players?

Sol :- From 6 players, 2 positions.

$$^6P_2 = \frac{6!}{}$$

$$= \frac{6 \times 5 \times \cancel{4} \times \cancel{3} \times \cancel{2} \times \cancel{1}}{\cancel{4} \times \cancel{3} \times \cancel{2} \times \cancel{1}}$$

$\Rightarrow 6 \times 5 \Rightarrow 30$

$\boxed{\text{Answer: } 30}$

4) How many different arrangements can be made using the letters of the word MATH?

Sol: MATH $\Rightarrow 4!$

$\Rightarrow 4 \times 3 \times 2 \times 1$

$\Rightarrow 12 \times 2 \times 1$

$\Rightarrow 24$

$\boxed{\text{Answer: } 24}$

5) A committee of 4 members is to be formed from 8 people. In how many ways can it be formed?

Sol:- we can select 4 members from committe to 8 people.

$$^8C_4 = \frac{8!}{4! \times 4!}$$

$$= \frac{8 \times 7 \times 6 \times 5 \times \cancel{4} \times \cancel{3} \times \cancel{2} \times \cancel{1}}{(\cancel{4} \times \cancel{3} \times \cancel{2} \times \cancel{1})\ (\cancel{4} \times \cancel{3} \times \cancel{2} \times \cancel{1})}$$

$$\Rightarrow \frac{\cancel{8}^{2} \times 7 \times \cancel{6}^{2} \times 5}{\cancel{4} \times \cancel{3} \times \cancel{2} \times 1}$$

$\Rightarrow 7 \times 5 \times 2$

$\Rightarrow 70$

$\boxed{\text{Answer: } 70}$

Block 2:

1) why are databases important in modern applications?

Databases important because, it used for store the data, retrieve the data and manage the data.

2) what does data consistency mean?

Data consistency means accurate, correct and same everywhere in the system.

) why do companies need well-designed database?

Companies need well-designed database because it improves performance, data consistency.

Grammar correction:

1) She do not understand SQL joins.

She does not understand SQL joins

2) He were explaining the database schema

He was explaining the database schema.

3) They has written the program successfully.

They have written the program successfully.

Block 3 - COMMUNICATION PRACTICE

One challenge is I faced while lose my coding practice during my semester. In the beginning, I was good at programming, but I stopped practicing to focus on my studies. After exams, I realized that I had forgotten many concepts and felt sad. I did not know where I want to start again. Then I decided to start slowly by practicing simple programs everyday. I learned from this lesson. Daily practice and consistency are very important.